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Check Your Knowledge: Optimization Techniques
Graded Assignment • 15 min

Due Nov 18, 11:59 PM IST

Your grade: 87.50%

Your latest: **87.50%** • Your highest: **87.50%** • To pass you need at least 80%. We keep your highest score.

Instructions

1. What is the primary goal of optimization in machine learning? 1 / 1 point

- Increasing the complexity of the model
- Maximizing the number of parameters
- Minimizing the error or loss of a model
- Reducing the size of the dataset

Nice work

Correct! The primary goal is to minimize the error or loss of a model.

2. Which type of optimization problem guarantees that any local minimum is also a global minimum? 1 / 1 point

- Convex optimization
- Stochastic optimization
- Constrained optimization
- Non-convex optimization

Nice work

Great job! Convex optimization guarantees that any local minimum is also a global minimum.

3. In gradient descent, what does the step-size (learning rate) control? 1 / 1 point

- The size of the steps taken towards the minimum
- The direction of the gradient
- The complexity of the objective function
- The number of iterations

Nice work

Correct! The step-size controls the size of the steps taken towards the minimum.

4. What is the main advantage of using Stochastic Gradient Descent (SGD) over traditional gradient descent? 1 / 1 point

- It always finds the global minimum
- It reduces computational cost by using a subset of data
- It requires fewer iterations
- It guarantees faster convergence

Nice work

Correct! SGD reduces computational cost by using a subset of data.

5. What is the purpose of Lagrange multipliers in constrained optimization? 1 / 1 point

- To convert a non-convex problem into a convex one
- To increase the number of constraints
- To simplify the objective function
- To transform a constrained problem into an unconstrained one

Nice work

Correct! Lagrange multipliers transform a constrained problem into an unconstrained one.

6. What does the convex conjugate (f^*(y)) represent? 1 / 1 point

- The integral of the function over its domain
- The maximum difference between the linear function (y^T x) and the original function (f(x))
- The sum of the function and its derivative
- The minimum value of the function

Nice work

Great job! The convex conjugate represents the maximum difference between the linear function (y^T x) and the original function (f(x)).

7. The convex conjugate of a convex function is always non-negative. 0 / 1 point

- false
- true

Try again

Not quite. Remember that the convex conjugate maintains the non-negativity property.

8. What is the involution property of the convex conjugate? 1 / 1 point

- It converts the function into a polynomial
- Applying the transform twice returns the original function
- It makes the function non-differentiable
- It ensures the function is always increasing

Nice work

Correct! The involution property means that applying the transform twice returns the original function.